

91

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QUANT CHECKLIST

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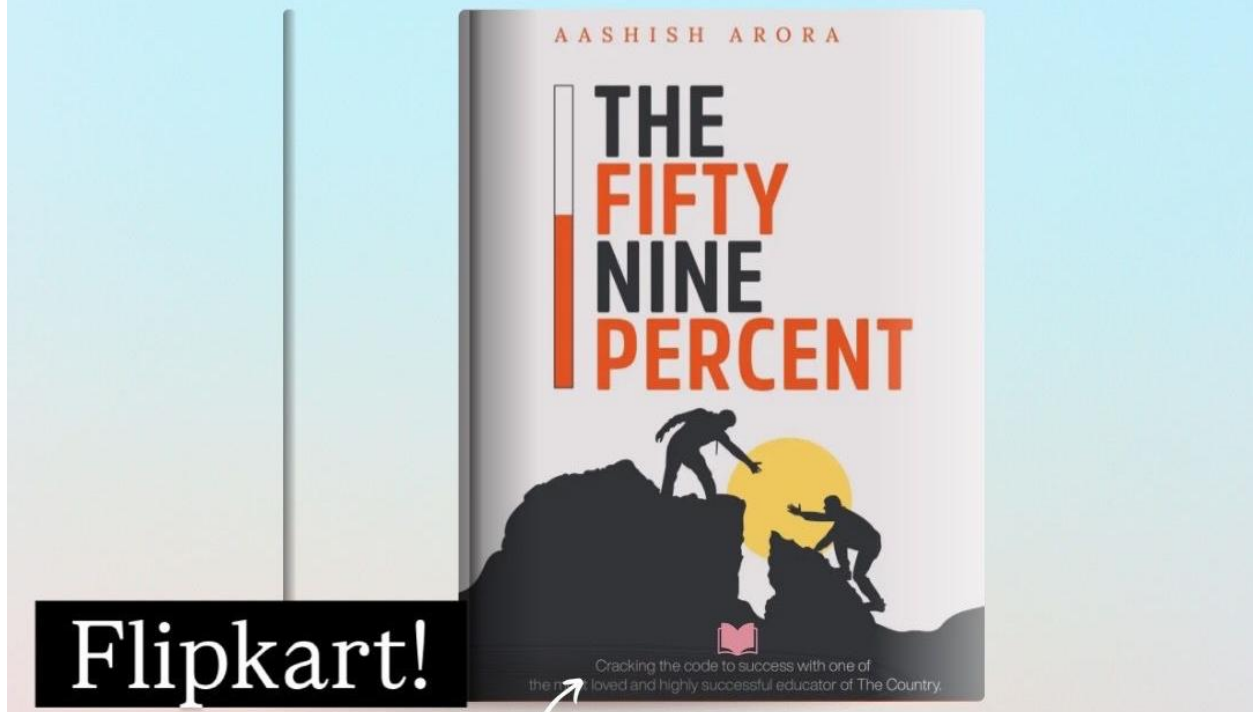


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CONTENTS

1. SIMPLIFICATION & APPROXIMATION	9
2. ARITHMETIC WORD PROBLEMS	22
3. QUADRATIC EQUATIONS	43
4. WRONG NUMBER SERIES	59
5. MISSING NUMBER SERIES	70
6. DATA INTERPRETATION	82

1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. 41.66% of $1296 + \frac{7}{12}$ of $900 - \sqrt{2209} = ? \times 4 + 157$

- A. 211.25
- B. 215.25
- C. 219.25
- D. 220.75
- E. None of these

2. $?%$ of $960 + 56.25\%$ of $384 = 37.5\%$ of $1152 + 18^2$

- A. 48.75
- B. 52.5
- C. 56.25
- D. 58.75
- E. None of these

3. $3\frac{2}{5}$ of $150 - 18.75\%$ of $640 + \sqrt{4096} = ? + 286$

- A. 156
- B. 160
- C. 164

D. 168

E. None of these

4. $(23^2 - 17^2) \div 6 + 27.27\%$ of $880 = ? \div 5$

- A. 1400
- B. 1420
- C. 1440
- D. 1460
- E. None of these

5. $\sqrt{?} + 38.46\%$ of $260 = 13^2 + 75$

- A. 19600
- B. 20164
- C. 20449
- D. 20736
- E. None of these

6. 83.33% of $540 + \frac{4}{9}$ of $675 - \sqrt[3]{91125} = ? \times 5$

- A. 139
- B. 141
- C. 143

D. 145

E. None of these

7. 125% of $576 - 72.72\%$ of $495 + 7^2 = ? - 18^3 \div 27$

A. 605

B. 615

C. 620

D. 625

E. None of these

8. $[14.28\%$ of $588 + 52\%$ of $800] \div 5 + 2^3 = ?$

A. 108

B. 112

C. 116

D. 120

E. None of these

9. $25 \times \sqrt{3844} - 16 \times \sqrt[3]{15625} + 31.25\%$ of $768 = ? \times 10$

A. 131

B. 135

C. 139

D. 143

E. None of these

10. 15.38% of $585 + \frac{9}{16}$ of $704 + 44.44\%$ of $729 = ?$

A. 800

B. 810

C. 820

D. 830

E. None of these

11. $? + 6.25\%$ of $4096 = (18 \times 37) - 14^2$

A. 202

B. 206

C. 210

D. 214

E. None of these

12. $\frac{8}{11}$ of $605 + \frac{11}{16}$ of $512 - \frac{5}{6}$ of $378 = ? \times 3$

A. 159

B. 163

C. 167

D. 171

E. None of these

13. $4\frac{1}{4}$ of $96 + 37.5\%$ of $528 - 12^2 = ? + 118$

A. 332

B. 338

C. 344

D. 350

E. None of these

14. $\sqrt{1764} + \sqrt[3]{110592} + \frac{7}{13}$ of 455 = ? $\times$
6 + 23

A. 50

B. 52

C. 54

D. 56

E. None of these

15. 64% of 1250 + $\frac{5}{8}$ of 736 - $\frac{3}{11}$ of
1089 = ? + 420

A. 543

B. 549

C. 555

D. 561

E. None of these

16. $(? \div 8) + 58.33\%$ of 864 = $4^3 \times 9 +$
96

A. 1284

B. 1324

C. 1344

D. 1384

E. None of these

17. 30.76% of 1547 + 18.75% of 1792
- 56.25% of 960 = ? $\times$ 8

A. 32

B. 34

C. 36

D. 38

E. None of these

18. $(52 \times 18 + 36^2 - 48 \times 17) \div 4 = ? +$
 $\sqrt{225}$

A. 327

B. 333

C. 336

D. 339

E. None of these

19. $\frac{2}{5}$ of 875 + $\frac{7}{9}$ of 756 - 62.5% of 448
= ?

A. 658

B. 662

C. 668

D. 672

E. None of these

20. 81.25% of 736 + $1\frac{7}{12}$ of 240 - $27^2 =$
? $\times$ 3

A. 79

- B. 83
C. 87
D. 91
E. None of these

Answers:

1. B
2. C
3. D
4. A
5. D
6. B
7. D
8. A
9. C
10. B
11. D
12. A
13. C
14. B
15. A
16. C
17. B
18. D
19. A
20. B

Solutions:

$$1. 41.66\% \text{ of } 1296 + \frac{7}{12} \text{ of } 900 - \sqrt{2209} \\ = ? \times 4 + 157$$

$$41.66\% = \frac{5}{12} \text{ and } \sqrt{2209} = 47$$

$$\frac{5}{12} \times 1296 + \frac{7}{12} \times 900 - 47 = ? \times 4 + 157$$

$$540 + 525 - 47 = ? \times 4 + 157$$

$$1018 = ? \times 4 + 157$$

$$? \times 4 = 861$$

$$? = 215.25$$

$$2. ?\% \text{ of } 960 + 56.25\% \text{ of } 384 = 37.5\% \\ \text{of } 1152 + 18^2$$

$$56.25\% = \frac{9}{16} \text{ and } 37.5\% = \frac{3}{8}$$

$$?\% \text{ of } 960 + \frac{9}{16} \times 384 = \frac{3}{8} \times 1152 + 324$$

$$?\% \text{ of } 960 + 216 = 432 + 324$$

$$?\% \text{ of } 960 = 756 - 216$$

$$?\% \text{ of } 960 = 540$$

$$? = 540 \times 100 \div 960 = 56.25$$

$$3. 3\frac{2}{5} \text{ of } 150 - 18.75\% \text{ of } 640 + \sqrt{4096} \\ = ? + 286$$

$$3\frac{2}{5} = \frac{17}{5}, 18.75\% = \frac{3}{16} \text{ and } \sqrt{4096} = 64$$

$$\frac{17}{5} \times 150 - \frac{3}{16} \times 640 + 64 = ? + 286$$

$$510 - 120 + 64 = ? + 286$$

$$454 = ? + 286$$

$$? = 168$$

$$4. (23^2 - 17^2) \div 6 + 27.27\% \text{ of } 880 = ? \div 5$$

$$27.27\% = \frac{3}{11}$$

$$(529 - 289) \div 6 + \frac{3}{11} \times 880 = ? \div 5$$

$$240 \div 6 + 240 = ? \div 5$$

$$280 = ? \div 5$$

$$? = 1400$$

$$5. \sqrt{?} + 38.46\% \text{ of } 260 = 13^2 + 75$$

$$38.46\% = \frac{5}{13} \text{ and } 13^2 = 169$$

$$\sqrt{?} + \frac{5}{13} \times 260 = 169 + 75$$

$$\sqrt{?} + 100 = 244$$

$$\sqrt{?} = 144$$

$$? = 144^2 = 20736$$

$$6. 83.33\% \text{ of } 540 + \frac{4}{9} \text{ of } 675 - \sqrt[3]{91125} = ? \times 5$$

$$83.33\% = \frac{5}{6} \text{ and } \sqrt[3]{91125} = 45$$

$$\frac{5}{6} \times 540 + \frac{4}{9} \times 675 - 45 = ? \times 5$$

$$450 + 300 - 45 = ? \times 5$$

$$705 = ? \times 5$$

$$? = 141$$

$$7. 125\% \text{ of } 576 - 72.72\% \text{ of } 495 + 7^2 = ? - 18^3 \div 27$$

$$125\% = \frac{5}{4}, 72.72\% = \frac{8}{11} \text{ and } 7^2 = 49$$

$$\frac{5}{4} \times 576 - \frac{8}{11} \times 495 + 49 = ? - 5832 \div 27$$

$$720 - 360 + 49 = ? - 216$$

$$409 = ? - 216$$

$$? = 625$$

$$8. [14.28\% \text{ of } 588 + 52\% \text{ of } 800] \div 5 + 2^3 = ?$$

$$14.28\% = \frac{1}{7} \text{ and } 2^3 = 8$$

$$[\frac{1}{7} \times 588 + 416] \div 5 + 8 = ?$$

$$[84 + 416] \div 5 + 8 = ?$$

$$500 \div 5 + 8 = ?$$

$$? = 108$$

$$9. 25 \times \sqrt{3844} - 16 \times \sqrt[3]{15625} + 31.25\% \text{ of } 768 = ? \times 10$$

$$\sqrt{3844} = 62, \sqrt[3]{15625} = 25 \text{ and } 31.25\% = \frac{5}{16}$$

$$25 \times 62 - 16 \times 25 + \frac{5}{16} \times 768 = ? \times 10$$

$$1550 - 400 + 240 = ? \times 10$$

$$1390 = ? \times 10$$

$$? = 139$$

$$10. 15.38\% \text{ of } 585 + \frac{9}{16} \text{ of } 704 + 44.44\% \text{ of } 729 = ?$$

$$15.38\% = \frac{2}{13} \text{ and } 44.44\% = \frac{4}{9}$$

$$\frac{2}{13} \times 585 + \frac{9}{16} \times 704 + \frac{4}{9} \times 729 = ?$$

$$90 + 396 + 324 = ?$$

$$? = 810$$

$$11. ? + 6.25\% \text{ of } 4096 = (18 \times 37) - 14^2$$

$$6.25\% = \frac{1}{16} \text{ and } 14^2 = 196$$

$$? + \frac{1}{16} \times 4096 = 666 - 196$$

$$? + 256 = 470$$

$$? = 214$$

$$12. \frac{8}{11} \text{ of } 605 + \frac{11}{16} \text{ of } 512 - \frac{5}{6} \text{ of } 378 = ? \times 3$$

$$\frac{8}{11} \times 605 + \frac{11}{16} \times 512 - \frac{5}{6} \times 378 = ? \times 3$$

$$440 + 352 - 315 = ? \times 3$$

$$477 = ? \times 3$$

$$? = 159$$

$$13. 4\frac{1}{4} \text{ of } 96 + 37.5\% \text{ of } 528 - 12^2 = ? + 118$$

$$4\frac{1}{4} = \frac{17}{4}, 37.5\% = \frac{3}{8} \text{ and } 12^2 = 144$$

$$\frac{17}{4} \times 96 + \frac{3}{8} \times 528 - 144 = ? + 118$$

$$408 + 198 - 144 = ? + 118$$

$$462 = ? + 118$$

$$? = 344$$

$$14. \sqrt{1764} + \sqrt[3]{110592} + \frac{7}{13} \text{ of } 455 = ? \times 6 + 23$$

$$\sqrt{1764} = 42 \text{ and } \sqrt[3]{110592} = 48$$

$$42 + 48 + \frac{7}{13} \times 455 = ? \times 6 + 23$$

$$42 + 48 + 245 = ? \times 6 + 23$$

$$335 = ? \times 6 + 23$$

$$? \times 6 = 312$$

$$? = 52$$

$$15. 64\% \text{ of } 1250 + \frac{5}{8} \text{ of } 736 - \frac{3}{11} \text{ of } 1089 = ? + 420$$

$$64\% \text{ of } 1250 = 800$$

$$\frac{5}{8} \times 736 = 460 \text{ and } \frac{3}{11} \times 1089 = 297$$

$$800 + 460 - 297 = ? + 420$$

$$963 = ? + 420$$

$$? = 543$$

$$16. (? \div 8) + 58.33\% \text{ of } 864 = 4^3 \times 9 + 96$$

$$58.33\% = \frac{7}{12} \text{ and } 4^3 = 64$$

$$? \div 8 + \frac{7}{12} \times 864 = 64 \times 9 + 96$$

$$? \div 8 + 504 = 576 + 96$$

$$? \div 8 = 672 - 504$$

$$? \div 8 = 168$$

$$? = 1344$$

$$17. 30.76\% \text{ of } 1547 + 18.75\% \text{ of } 1792 - 56.25\% \text{ of } 960 = ? \times 8$$

$$30.76\% = \frac{4}{13}, 18.75\% = \frac{3}{16} \text{ and } 56.25\% = \frac{9}{16}$$

$$\frac{4}{13} \times 1547 + \frac{3}{16} \times 1792 - \frac{9}{16} \times 960 = ? \times 8$$

$$476 + 336 - 540 = ? \times 8$$

$$272 = ? \times 8$$

$$? = 34$$

$$18. (52 \times 18 + 36^2 - 48 \times 17) \div 4 = ? + \sqrt{225}$$

$$36^2 = 1296 \text{ and } \sqrt{225} = 15$$

$$(936 + 1296 - 816) \div 4 = ? + 15$$

$$1416 \div 4 = ? + 15$$

$$354 = ? + 15$$

$$? = 339$$

$$19. \frac{2}{5} \text{ of } 875 + \frac{7}{9} \text{ of } 756 - 62.5\% \text{ of } 448 = ?$$

$$62.5\% = \frac{5}{8}$$

$$\frac{2}{5} \times 875 + \frac{7}{9} \times 756 - \frac{5}{8} \times 448 = ?$$

$$350 + 588 - 280 = ?$$

$$? = 658$$

$$20. 81.25\% \text{ of } 736 + 1\frac{7}{12} \text{ of } 240 - 27^2 = ? \times 3$$

$$81.25\% = \frac{13}{16}, 1\frac{7}{12} = \frac{19}{12} \text{ and } 27^2 = 729$$

$$\frac{13}{16} \times 736 + \frac{19}{12} \times 240 - 729 = ? \times 3$$

$$598 + 380 - 729 = ? \times 3$$

$$249 = ? \times 3$$

$$? = 83$$

2. ARITHMETIC QUESTIONS

Q1: A cylinder and a sphere have the same radius. The ratio of the volume of the cylinder to the volume of the sphere is 3:2. If the ratio of their total surface areas is $a:b$, find $a + b$.

एक बेलन और एक गोले की त्रिज्या समान है। बेलन के आयतन और गोले के आयतन का अनुपात 3:2 है। यदि उनके कुल पृष्ठीय क्षेत्रफलों का अनुपात $a:b$ है, तो $a + b$ ज्ञात कीजिए।

- (A) 7
- (B) 3
- (C) 5
- (D) 8
- (E) 6

Q2: The difference between the circumferences of two circles A and B is 176 cm. The radius of circle B is 5 times the radius of circle A. Find the area of circle B.

Take $\pi = 22/7$.

दो वृत्त A और B की परिधियों के बीच का अंतर 176 सेमी है। वृत्त B की त्रिज्या, वृत्त A की त्रिज्या की 5 गुना है। वृत्त B का क्षेत्रफल ज्ञात कीजिए। $\pi = 22/7$ लीजिए।

- (A) 4070
- (B) 3520
- (C) 4620
- (D) 3850
- (E) 4200

Q3: A shopkeeper marks an item 40% above its cost price and earns Rs. 150 after giving a 25% discount on the marked price. If he gives only a 10% discount on the marked price, find his profit on one item.

एक दुकानदार किसी वस्तु का अंकित मूल्य उसके क्रय मूल्य से 40% अधिक रखता है और अंकित मूल्य पर 25% छूट देने के बाद 150 रुपये का लाभ कमाता है। यदि वह अंकित मूल्य पर केवल 10% छूट देता है, तो एक वस्तु पर उसका लाभ ज्ञात कीजिए।

- (A) 720
- (B) 780
- (C) 840
- (D) 690

(E) 900

Q4: The average weight of N boys in a class is 42 kg, and the average weight of 20 girls in the same class is 30 kg. If the total weight of all students is 1692 kg, find the value of $1.5N$.

एक कक्षा में N लड़कों का औसत वजन 42 किग्रा है और उसी कक्षा की 20 लड़कियों का औसत वजन 30 किग्रा है। यदि सभी विद्यार्थियों का कुल वजन 1692 किग्रा है, तो $1.5N$ का मान ज्ञात कीजिए।

- (A) 45
- (B) 33
- (C) 42
- (D) 36
- (E) 39

Q5: Kavita invests Rs. X in a scheme that offers simple interest at 15% per annum for 3 years and receives Rs. 2700 as interest. If she invests Rs. $(X + 1000)$ in another scheme that offers compound interest at 20% per annum for 2 years, compounded annually, find the compound interest received by her.

कविता Rs. X को एक योजना में 3 वर्षों के लिए 15% वार्षिक साधारण ब्याज पर निवेश करती है और उसे 2700 रुपये ब्याज मिलता है। यदि वह Rs. $(X + 1000)$ को दूसरी योजना में 2 वर्षों के लिए 20% वार्षिक चक्रवृद्धि ब्याज पर निवेश करती है, जो वार्षिक रूप से संयोजित होता है, तो उसे प्राप्त चक्रवृद्धि ब्याज ज्ञात कीजिए।

- (A) Rs. 3080
- (B) Rs. 3180
- (C) Rs. 2960
- (D) Rs. 3360
- (E) Rs. 2880

Q6: Two trains P and Q are 360 meters and $(360 + x)$ meters long, respectively. Both trains run at the same speed. Train P and train Q cross a pole in 12 seconds and 20 seconds, respectively. Find the time taken by train Q to cross a 300-meter-long platform.

दो ट्रेन P और Q की लंबाई क्रमशः 360 मीटर और $(360 + x)$ मीटर है। दोनों ट्रेन समान गति से चलती हैं। ट्रेन P और ट्रेन Q एक खंभे को क्रमशः 12 सेकंड और 20 सेकंड में पार करती हैं। ट्रेन Q को 300 मीटर लंबे प्लेटफॉर्म को पार करने में कितना समय लगेगा?

- (A) 25 sec
- (B) 30 sec

- (C) 35 sec
- (D) 40 sec
- (E) 45 sec

Q7: Two partners invest in a business in the ratio of 4:5. The first partner receives Rs. 5600 as profit out of the total profit of Rs. 9800. If the difference between their investment periods is 4 months, then find for how many months the first partner kept his investment.

दो साझेदार एक व्यवसाय में 4:5 के अनुपात में निवेश करते हैं। पहले साझेदार को Rs. 9800 के कुल लाभ में से Rs. 5600 लाभ के रूप में प्राप्त होते हैं। यदि उनके निवेश की समयावधि के बीच 4 महीने का अंतर है, तो पहले साझेदार ने अपना निवेश कितने महीनों तक रखा?

- (A) 12
- (B) 6
- (C) 15
- (D) 10
- (E) 8

Q8: The still-water speeds of boat P and boat Q are in the ratio of 7:5, respectively. The speed of the stream is 8 km/hr. Boat P covers 200 km downstream in 4 hours. Find the time taken by boat Q to cover 176 km upstream.

नाव P और नाव Q की शांत जल में गति का अनुपात क्रमशः 7:5 है। धारा की गति 8 किमी/घंटा है। नाव P धारा के अनुकूल 200 किमी की दूरी 4 घंटे में तय करती है। नाव Q को धारा के प्रतिकूल 176 किमी की दूरी तय करने में कितना समय लगेगा?

- (A) 8 hours
- (B) 10 hours
- (C) 12 hours
- (D) 16 hours
- (E) 18 hours

Q9: The present age of Nisha is 30 years. Six years later, the ratio of Kunal's age to Nisha's age will be 7:9. If Varun's present age is 50% more than Kunal's present age, then find Varun's age five years ago.

निशा की वर्तमान आयु 30 वर्ष है। छह वर्ष बाद, कुणाल और निशा की आयु का अनुपात 7:9 होगा। यदि वरुण की वर्तमान आयु कुणाल की वर्तमान आयु से 50% अधिक है, तो पांच वर्ष पहले वरुण की आयु ज्ञात कीजिए।

- (A) 24 years
- (B) 26 years

- (C) 30 years
- (D) 32 years
- (E) 28 years

Q10: A person spends 18% of his monthly salary on food and 12% on rent. He invests Rs. 14000 in a recurring deposit and saves the remaining amount. If his total monthly salary is Rs. 40000 and his savings are $5x\%$ of his salary, then find x .
 एक व्यक्ति अपनी मासिक आय का 18% भोजन पर और 12% किराए पर खर्च करता है। वह Rs. 14000 एक आवर्ती जमा में निवेश करता है और शेष राशि बचाता है। यदि उसकी कुल मासिक आय Rs. 40000 है और उसकी बचत उसके वेतन का $5x\%$ है, तो x ज्ञात कीजिए।

- (A) 5
- (B) 6
- (C) 7
- (D) 8
- (E) 10

Q11: A dairy can has 900 ml of a milk-water mixture in which milk is 55%. How much pure milk should be poured into the can so that milk becomes 70% of the final mixture?

एक डेयरी कैन में 900 मिली दूध और पानी का मिश्रण है, जिसमें दूध 55% है। कैन में कितना शुद्ध दूध मिलाया जाए ताकि अंतिम मिश्रण में दूध 70% हो जाए?

- (A) 360 ml
- (B) 400 ml
- (C) 500 ml
- (D) 520 ml
- (E) 450 ml

Q12: Tap M can fill a tank in 36 minutes and Tap N can fill the same tank in 54 minutes. Both taps are opened together. After how many minutes should Tap M be closed so that Tap N alone fills the remaining tank in 9 minutes?

नल M एक टंकी को 36 मिनट में भर सकता है और नल N उसी टंकी को 54 मिनट में भर सकता है। दोनों नल एक साथ खोले जाते हैं। कितने मिनट बाद नल M को बंद कर देना चाहिए ताकि बची हुई टंकी को नल N अकेले 9 मिनट में भर दे?

- (A) 12 minutes
- (B) 15 minutes
- (C) 18 minutes
- (D) 21 minutes

(E) 24 minutes

Q13: Partner P and Partner Q started a venture with investments of Rs 1500 and Rs 2000, respectively. After 6 months, P increased his investment by 60%, while Q reduced his investment by 25%. If the total profit at the end of the year was Rs 8880, find P's profit share.

साझेदार P और साझेदार Q ने क्रमशः Rs 1500 और Rs 2000 के निवेश से एक व्यवसाय शुरू किया। 6 महीने बाद, P ने अपना निवेश 60% बढ़ा दिया, जबकि Q ने अपना निवेश 25% घटा दिया। यदि वर्ष के अंत में कुल लाभ Rs 8880 था, तो P का लाभांश ज्ञात कीजिए।

- (A) Rs 4200
- (B) Rs 4440
- (C) Rs 4560
- (D) Rs 4680
- (E) Rs 4920

Q14: A delivery van covers $\frac{1}{5}$ of a route at 30 km/hr, then covers $\frac{1}{2}$ of the remaining route at 45 km/hr, and covers the rest at 36 km/hr. If the total time taken for the whole journey is 8 hours, find the total length of the route.

एक डिलीवरी वैन किसी मार्ग का $\frac{1}{5}$ भाग 30 किमी/घंटा की गति से तय करती है, फिर बचे हुए मार्ग का $\frac{1}{2}$ भाग 45 किमी/घंटा की गति से तय करती है, और बाकी मार्ग 36 किमी/घंटा की गति से तय करती है। यदि पूरी यात्रा में कुल 8 घंटे लगते हैं, तो मार्ग की कुल लंबाई ज्ञात कीजिए।

- (A) 300 km
- (B) 250 km
- (C) 275 km
- (D) 325 km
- (E) 350 km

Q15: A right-angled triangular sheet has a base of 14 cm and a height of 48 cm. It is rotated about its height to form a cone. Find the total surface area of the cone. Use $\pi = \frac{22}{7}$.

एक समकोण त्रिभुजाकार शीट का आधार 14 सेमी और ऊंचाई 48 सेमी है। इसे इसकी ऊंचाई के चारों ओर घुमाकर एक शंकु बनाया जाता है। शंकु का कुल पृष्ठीय क्षेत्रफल ज्ञात कीजिए। $\pi = \frac{22}{7}$ का प्रयोग कीजिए।

- (A) 2688 cm²
- (B) 2816 cm²
- (C) 2944 cm²

- (D) 3024 cm^2
 (E) 2464 cm^2

Q16: The average present age of P and Q is 33 years. The present age of R is 50% more than the present age of Q. If the sum of the present ages of P, Q and R is 105 years, then find the present age of P.

P और Q की वर्तमान आयु का औसत 33 वर्ष है। R की वर्तमान आयु, Q की वर्तमान आयु से 50% अधिक है। यदि P, Q और R की वर्तमान आयु का योग 105 वर्ष है, तो P की वर्तमान आयु ज्ञात कीजिए।

- (A) 36 years
 (B) 44 years
 (C) 46 years
 (D) 40 years
 (E) 52 years

Q17: Three numbers are in the ratio 5:8:13. The sum of the smallest and the largest number exceeds the sum of the smallest and the second largest number by 45. Find the largest number.

तीन संख्याएँ 5:8:13 के अनुपात में हैं। सबसे छोटी और सबसे बड़ी संख्या का योग, सबसे छोटी और दूसरी सबसे बड़ी संख्या के योग से 45 अधिक है। सबसे बड़ी संख्या ज्ञात कीजिए।

- (A) 108
 (B) 117
 (C) 126
 (D) 99
 (E) 135

Q18: If the numerator of a fraction is increased by 25% and its denominator is decreased by 20%, then the new fraction exceeds the original fraction by $\frac{9}{20}$. Find the sum of the numerator and denominator of the original fraction, if the fraction is in its simplest form.

यदि किसी भिन्न के अंश में 25% की वृद्धि की जाती है और उसके हर में 20% की कमी की जाती है, तो नया भिन्न मूल भिन्न से $\frac{9}{20}$ अधिक हो जाता है। यदि मूल भिन्न अपने सरलतम रूप में है, तो उसके अंश और हर का योग ज्ञात कीजिए।

- (A) 8
 (B) 11
 (C) 9
 (D) 7

(E) 13

Q19: A person invests 30% of Rs. x at 20% p.a. compound interest, compounded annually. If the interest earned after 2 years is Rs. 3960, then find the value of x .

एक व्यक्ति x रुपये का 30% भाग 20% वार्षिक चक्रवृद्धि ब्याज पर निवेश करता है, जो वार्षिक रूप से संयोजित होता है। यदि 2 वर्षों के बाद अर्जित ब्याज 3960 रुपये है, तो x का मान ज्ञात कीजिए।

(A) Rs. 32000

(B) Rs. 24000

(C) Rs. 28000

(D) Rs. 36000

(E) Rs. 30000

Q20: A trader bought two articles for Rs. 5000. He sold one article at a profit of 25% and the other article at a loss of 15%. If his overall profit on selling both articles is 7%, then find the cost price of the article sold at a loss.

एक व्यापारी ने दो वस्तुएँ 5000 रुपये में खरीदीं। उसने एक वस्तु को 25% लाभ पर और दूसरी वस्तु को 15% हानि पर बेचा। यदि दोनों वस्तुओं को बेचने पर उसका कुल लाभ 7% है, तो हानि पर बेची गई वस्तु का क्रय मूल्य ज्ञात कीजिए।

(A) Rs. 2250

(B) Rs. 2500

(C) Rs. 2000

(D) Rs. 2750

(E) Rs. 2400

Answers:

1. C

2. D

3. B

4. E

5. A

6. B

7. D

8. A

9. E

10. C

11. E

12. C

13. D

14. A

15. B

16. D

17. B

18. C

19. E

20. A

Solutions:**1. Option C**

Let the common radius be r and height of the cylinder be h .

Volume of cylinder = $\pi \times r \times r \times h$

Volume of sphere = $\frac{4}{3} \times \pi \times r \times r \times r$

Given ratio = 3:2

So, $\pi r^2 h : \frac{4}{3} \pi r^3 = 3:2$

This means $3h : 4r = 3:2$

So, $3h/4r = 3/2$

$6h = 12r$

$h = 2r$

Total surface area of cylinder = $2\pi r(r + h)$

Total surface area of cylinder = $2\pi r(r + 2r) = 6\pi r^2$

Total surface area of sphere = $4\pi r^2$

Ratio of total surface area of cylinder to sphere = $6\pi r^2 : 4\pi r^2 = 3:2$

So, $a = 3$ and $b = 2$

$a + b = 3 + 2 = 5$

2. Option D

Let the radius of circle A be r cm.

Radius of circle B = $5r$ cm

Difference between circumferences = 176 cm

Circumference of circle = $2\pi r$

So, $2\pi \times 5r - 2\pi \times r = 176$

$10\pi r - 2\pi r = 176$

$8\pi r = 176$

Using $\pi = 22/7$

$8 \times 22/7 \times r = 176$

$176r/7 = 176$

$r = 7$ cm

Radius of circle B = $5 \times 7 = 35$ cm

Area of circle B = πr^2

$$\text{Area} = 22/7 \times 35 \times 35$$

$$\text{Area} = 3850 \text{ sq cm}$$

3. Option B

Let the cost price be 100 units.

Marked price = 140% of cost price = 140 units

Selling price after 25% discount = $140 \times 75/100 = 105$ units

Profit = $105 - 100 = 5$ units

Given profit = Rs. 150

So, 5 units = Rs. 150

1 unit = $150/5 = \text{Rs. } 30$

Cost price = $100 \times 30 = \text{Rs. } 3000$

Marked price = $140 \times 30 = \text{Rs. } 4200$

If discount is 10%, then selling price = $4200 \times 90/100$

Selling price = Rs. 3780

Profit = $3780 - 3000 = \text{Rs. } 780$

4. Option E

Average weight of 20 girls = 30 kg

Total weight of girls = $20 \times 30 = 600$ kg

Total weight of all students = 1692 kg

Total weight of boys = $1692 - 600 = 1092$ kg

Average weight of N boys = 42 kg

So, $N = 1092/42 = 26$

Required value = 1.5N

$1.5N = 1.5 \times 26 = 39$

5. Option A

Simple interest = Rs. 2700

Rate = 15% per annum

Time = 3 years

Simple interest = $\text{Principal} \times \text{Rate} \times \text{Time}/100$

$$2700 = X \times 15 \times 3/100$$

$$2700 = 45X/100$$

$$X = 2700 \times 100/45$$

$$X = \text{Rs. } 6000$$

Amount invested in second scheme = $X + 1000$

Amount invested = $6000 + 1000 = \text{Rs. } 7000$

For 2 years at 20% compound interest, effective rate = $20 + 20 + 20 \times 20/100$

Effective rate = $40 + 4 = 44\%$

Compound interest = $7000 \times 44/100$

Compound interest = Rs. 3080

6. Option B

Train P crosses a pole, so distance covered = length of train P = 360 meters.

Speed of train P = $360 \div 12 = 30 \text{ m/sec.}$

Both trains have the same speed, so speed of train Q = 30 m/sec.

Train Q crosses a pole in 20 seconds.

Length of train Q = $30 \times 20 = 600 \text{ meters.}$

To cross a 300-meter platform, total distance covered by train Q = $600 + 300 = 900 \text{ meters.}$

Time taken = $900 \div 30 = 30 \text{ seconds.}$

7. Option D

Profit of first partner = Rs. 5600.

Profit of second partner = $9800 - 5600 = \text{Rs. } 4200.$

Profit ratio = $5600:4200 = 4:3.$

Investment ratio = 4:5.

Profit ratio = Investment ratio $\times$ Time ratio.

So, time ratio = Profit ratio $\div$ Investment ratio.

Time ratio = $4/3 \div 4/5 = 4/3 \times 5/4 = 5/3.$

So, time period ratio of first partner to second partner = 5:3.

Difference = $5 - 3 = 2 \text{ parts.}$

2 parts = 4 months.

1 part = 2 months.

Time period of first partner = $5 \times 2 = 10 \text{ months.}$

8. Option A

Boat P covers 200 km downstream in 4 hours.

Downstream speed of boat P = $200 \div 4 = 50 \text{ km/hr.}$

Speed of stream = 8 km/hr.

Still-water speed of boat P = $50 - 8 = 42 \text{ km/hr.}$

Still-water speed ratio of boat P and boat Q = 7:5.

7 parts = 42 km/hr.

1 part = $42 \div 7 = 6$ km/hr.

Still-water speed of boat Q = $5 \times 6 = 30$ km/hr.

Upstream speed of boat Q = $30 - 8 = 22$ km/hr.

Time taken by boat Q to cover 176 km upstream = $176 \div 22 = 8$ hours.

9. Option E

Present age of Nisha = 30 years.

After 6 years, Nisha's age = $30 + 6 = 36$ years.

After 6 years, Kunal:Nisha = 7:9.

9 parts = 36 years.

1 part = $36 \div 9 = 4$ years.

Kunal's age after 6 years = $7 \times 4 = 28$ years.

Present age of Kunal = $28 - 6 = 22$ years.

Varun's present age is 50% more than Kunal's present age.

Varun's present age = $22 + 50\% \text{ of } 22 = 22 + 11 = 33$ years.

Varun's age five years ago = $33 - 5 = 28$ years.

10. Option C

Monthly salary = Rs. 40000.

Amount spent on food = 18% of 40000 = 7200.

Amount spent on rent = 12% of 40000 = 4800.

Amount invested = Rs. 14000.

Total amount used = $7200 + 4800 + 14000 = \text{Rs. } 26000$.

Savings = $40000 - 26000 = \text{Rs. } 14000$.

Savings percent = $14000/40000 \times 100 = 35\%$.

Given savings = 5x% of salary.

So, $5x = 35$.

$x = 35 \div 5 = 7$.

11. Option E

Initial mixture = 900 ml.

Milk = 55% of 900 = 495 ml.

Water = $900 - 495 = 405$ ml.

Only pure milk is added, so water remains 405 ml.

In the final mixture, milk is 70%, so water is 30%.

30% of final mixture = 405.

Final mixture = $405 \times 100/30 = 1350$ ml.

Milk added = $1350 - 900 = 450$ ml.

12. Option C

Take total tank capacity = 108 units.

Efficiency of Tap M = $108 \div 36 = 3$ units/minute.

Efficiency of Tap N = $108 \div 54 = 2$ units/minute.

Tap N alone fills the remaining tank in 9 minutes.

Remaining work = $2 \times 9 = 18$ units.

Work done by both taps together = $108 - 18 = 90$ units.

Combined efficiency of M and N = $3 + 2 = 5$ units/minute.

Time for which both taps were opened = $90 \div 5 = 18$ minutes.

13. Option D

Profit ratio is based on investment $\times$ time.

For Partner P:

Initial investment for 6 months = $1500 \times 6 = 9000$.

After 60% increase, new investment = $1500 + 900 = 2400$.

Investment for next 6 months = $2400 \times 6 = 14400$.

Total capital-time of P = $9000 + 14400 = 23400$.

For Partner Q:

Initial investment for 6 months = $2000 \times 6 = 12000$.

After 25% reduction, new investment = $2000 - 500 = 1500$.

Investment for next 6 months = $1500 \times 6 = 9000$.

Total capital-time of Q = $12000 + 9000 = 21000$.

Profit ratio of P and Q = $23400:21000 = 39:35$.

Total ratio = $39 + 35 = 74$.

P's share = $8880 \times 39/74 = \text{Rs } 4680$.

14. Option A

Let the total route be D km.

Distance covered in first part = $D/5$.

Time for first part = $D/5 \div 30 = D/150$ hours.

Remaining distance = $4D/5$.

Half of remaining distance = $2D/5$.

Time for second part = $2D/5 \div 45 = 2D/225$ hours.

Remaining last distance = $2D/5$.

Time for last part = $2D/5 \div 36 = D/90$ hours.

Total time = $D/150 + 2D/225 + D/90$.

LCM of 150, 225 and 90 = 450.

Total time = $3D/450 + 4D/450 + 5D/450 = 12D/450 = 2D/75$.

Given total time = 8 hours.

$2D/75 = 8$.

$D = 8 \times 75 \div 2 = 300$ km.

15. Option B

When the right-angled triangle is rotated about its height, a cone is formed.

Radius of cone = base of triangle = 14 cm.

Height of cone = 48 cm.

Slant height = square root of $14 \times 14 + 48 \times 48$.

Slant height = square root of $196 + 2304$ = square root of 2500 = 50 cm.

Total surface area of cone = $\pi \times r \times (r + l)$.

TSA = $22/7 \times 14 \times (14 + 50)$.

TSA = $22/7 \times 14 \times 64$.

TSA = $44 \times 64 = 2816$ cm²

16. Option D

Average age of P and Q = 33 years

Sum of ages of P and Q = $33 \times 2 = 66$ years

Let the age of Q = x years.

Age of R = 50% more than Q = $x + 50\% \text{ of } x = 150x/100 = 3x/2$

Sum of ages of P, Q and R = 105 years

So, $66 + 3x/2 = 105$

$3x/2 = 105 - 66 = 39$

$3x = 78$

$x = 26$

Age of Q = 26 years

Age of P = $66 - 26 = 40$ years

17. Option B

Let the three numbers be 5x, 8x and 13x.

Sum of smallest and largest number = $5x + 13x = 18x$

Sum of smallest and second largest number = $5x + 8x = 13x$

$$\text{Difference} = 18x - 13x = 5x$$

$$\text{Given difference} = 45$$

$$5x = 45$$

$$x = 9$$

$$\text{Largest number} = 13x = 13 \times 9 = 117$$

18. Option C

Let the original fraction be a/b .

Numerator is increased by 25%, so new numerator = $125a/100 = 5a/4$

Denominator is decreased by 20%, so new denominator = $80b/100 = 4b/5$

$$\text{New fraction} = (5a/4) / (4b/5)$$

$$\text{New fraction} = 5a/4 \times 5/4b = 25a/16b$$

So, new fraction = $25/16$ of the original fraction.

$$\text{Difference between new fraction and original fraction} = 25a/16b - a/b$$

$$\text{Difference} = 25a/16b - 16a/16b = 9a/16b$$

$$\text{Given difference} = 9/20$$

$$\text{So, } 9a/16b = 9/20$$

$$a/b = 16/20 = 4/5$$

$$\text{Original fraction} = 4/5$$

$$\text{Sum of numerator and denominator} = 4 + 5 = 9$$

19. Option E

$$\text{Amount invested} = 30\% \text{ of } x = 30x/100$$

$$\text{Rate} = 20\% \text{ p.a.}$$

$$\text{Time} = 2 \text{ years}$$

Shortcut for compound interest percentage for 2 years at 20%:

$$20 + 20 + 20 \times 20/100 = 44\%$$

So, compound interest = 44% of the invested amount

$$\text{Given interest} = \text{Rs. } 3960$$

$$3960 = 44\% \text{ of } 30\% \text{ of } x$$

$$3960 = 44/100 \times 30/100 \times x$$

$$3960 = 1320x/10000$$

$$3960 = 13.2\% \text{ of } x$$

$$x = 3960 \times 100/13.2$$

$$x = 30000$$

20. Option A

One article is sold at 25% profit.

Other article is sold at 15% loss.

Overall profit = 7%

Using alligation:

Profit article : Loss article = Difference from mean profit on loss side : Difference from mean profit on profit side

Profit article : Loss article = $7 - (-15) : 25 - 7$

Profit article : Loss article = 22 : 18

Profit article : Loss article = 11 : 9

Total cost price = Rs. 5000

Cost price of article sold at loss = $9/(11 + 9) \times 5000$

Cost price of article sold at loss = $9/20 \times 5000$

Cost price of article sold at loss = Rs. 2250

3. Quadratic Equations

CHECKLIST

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

A. if $x > y$

B. if $x \geq y$

C. if $x < y$

D. if $x \leq y$

E. if $x = y$ or relationship cannot be established

1.) I. $10x^2 - 35x + 30 = 0$
II. $12y^2 - 10y - 2 = 0$

2.) I. $x^2 - 7\sqrt{10}x + 60 = 0$
II. $y^2 - 5\sqrt{10}y + 40 = 0$

3.) I. $12x^2 + 17x - 5 = 0$
II. $6y^2 - 61y + 154 = 0$

4.) I. $9x^2 + 42x + 48 = 0$
II. $4y^2 - 32y - 80 = 0$

5.) I. $x^2 + 7\sqrt{2}x + 24 = 0$
II. $y^2 + 3\sqrt{2}y + 4 = 0$

6.) I. $5x^2 + 5x - 10 = 0$
II. $2y^2 + 12y + 16 = 0$

7.) I. $6x^2 - 38x + 60 = 0$
II. $4y^2 - 64 = 0$

8.) I. $x^2 + 6\sqrt{3}x + 15 = 0$
II. $y^2 + 2\sqrt{3}y + 3 = 0$

9.) I. $12x^2 - 76x + 80 = 0$
II. $8y^2 + 26y + 6 = 0$

10.) I. $4x^2 + 14x - 18 = 0$
II. $6y^2 - 47y + 66 = 0$

11.) I. $6x^2 - 37x - 35 = 0$
 II. $3y^2 + 7y - 40 = 0$

12.) I. $x^2 - 4\sqrt{13}x + 39 = 0$
 II. $y^2 - 7\sqrt{13}y + 78 = 0$

13.) I. $15x^2 + 50x + 40 = 0$
 II. $8y^2 + 76y + 120 = 0$

14.) I. $6x^2 - 49x + 8 = 0$
 II. $10y^2 + 54y + 20 = 0$

15.) I. $8x^2 - 22x + 15 = 0$
 II. $4y^2 - 50y + 66 = 0$

16.) I. $x^2 + 3\sqrt{5}x + 10 = 0$
 II. $y^2 + 7\sqrt{5}y + 60 = 0$

17.) I. $4x^2 + 36x + 80 = 0$
 II. $12y^2 - 17y - 44 = 0$

18.) I. $10x^2 - 74x - 48 = 0$
 II. $15y^2 - 66y - 144 = 0$

19.) I. $x^2 + 2\sqrt{2}x + 2 = 0$
 II. $y^2 + 6\sqrt{2}y + 10 = 0$

20.) I. $3x^2 + 39x + 36 = 0$
 II. $6y^2 - 54y - 60 = 0$

Answers:

1. A
2. E
3. C
4. D
5. C
6. B
7. E
8. D
9. A
10. C
11. E
12. E
13. B
14. A
15. D
16. A
17. C
18. E
19. B
20. D

Solutions:

(1) $x = \frac{3}{2}, 2$

$y = 1, -\frac{1}{6}$

(2) $x = \sqrt{10}, 6\sqrt{10}$

$y = \sqrt{10}, 4\sqrt{10}$

$$(3) x = -\frac{5}{3}, \frac{1}{4}$$

$$y = \frac{11}{2}, \frac{14}{3}$$

$$(4) x = -\frac{8}{3}, -2$$

$$y = -2, 10$$

$$(5) x = -3\sqrt{2}, -4\sqrt{2}$$

$$y = -\sqrt{2}, -2\sqrt{2}$$

$$(6) x = 1, -2$$

$$y = -2, -4$$

$$(7) x = \frac{10}{3}, 3$$

$$y = 4, -4$$

$$(8) x = -\sqrt{3}, -5\sqrt{3}$$

$$y = -\sqrt{3}, -\sqrt{3}$$

$$(9) x = 5, \frac{4}{3}$$

$$y = -3, -\frac{1}{4}$$

$$(10) x = 1, -\frac{9}{2}$$

$$y = \frac{11}{6}, 6$$

$$(11) x = 7, -\frac{5}{6}$$

$$y = -5, \frac{8}{3}$$

$$(12) x = \sqrt{13}, 3\sqrt{13}$$

$$y = \sqrt{13}, 6\sqrt{13}$$

$$(13) x = -\frac{4}{3}, -2$$

$$y = -\frac{15}{2}, -2$$

$$(14) x = 8, \frac{1}{6}$$

$$y = -5, -\frac{2}{5}$$

$$(15) x = \frac{5}{4}, \frac{3}{2}$$

$$y = \frac{3}{2}, 11$$

$$(16) x = -\sqrt{5}, -2\sqrt{5}$$

$$y = -3\sqrt{5}, -4\sqrt{5}$$

$$(17) x = -5, -4$$

$$y = -\frac{4}{3}, \frac{11}{4}$$

$$(18) x = 8, -\frac{3}{5}$$

$$y = 6, -\frac{8}{5}$$

$$(19) x = -\sqrt{2}, -\sqrt{2}$$

$$y = -\sqrt{2}, -5\sqrt{2}$$

$$(20) x = -12, -1$$

$$y = -1, 10$$

4. WRONG NUMBER SERIES

(1) 1540, 1578, 1631, 1701, 1783, 1900

- (a) 1783
- (b) 1631
- (c) 1900
- (d) 1578
- (e) None of these

(2) 5200, 5140, 5020, 4822, 4474, 3970

- (a) 3970
- (b) 4822
- (c) 5140
- (d) 4474
- (e) None of these

(3) 214, 431, 876, 1746, 3507, 7035

- (a) 431
- (b) 3507
- (c) 876
- (d) 7035
- (e) None of these

(4) 34652, 17337, 5792, 1465, 312, 82

- (a) 5792
- (b) 312
- (c) 17337
- (d) 82
- (e) None of these

(5) 860, 904, 956, 1024, 1128, 1240

- (a) 904
- (b) 956
- (c) 1024
- (d) 1240
- (e) None of these

(6) 142, 298, 915, 3698, 18475, 110892

- (a) 3698
- (b) 915
- (c) 18475
- (d) 110892
- (e) None of these

(7) 2450, 2414, 2368, 2281, 2176, 2040

- (a) 2176

- | | |
|---|---|
| (b) 2368 | (11) 1260, 1278, 1318, 1388, 1496, 1662 |
| (c) 2450 | (a) 1662 |
| (d) 2281 | (b) 1388 |
| (e) None of these | (c) 1318 |
| | (d) 1260 |
| (8) 312.75, 319, 331.5, 356.5, 416.5, 506.5 | (e) None of these |
| (a) 319 | |
| (b) 506.5 | (12) 49575, 16530, 5535, 1860, 660, 300 |
| (c) 416.5 | (a) 16530 |
| (d) 356.5 | (b) 5535 |
| (e) None of these | (c) 660 |
| | (d) 300 |
| (9) 533, 584, 637, 701, 749, 808 | (e) None of these |
| (a) 584 | |
| (b) 808 | (13) 393, 427, 463, 501, 552, 583 |
| (c) 637 | (a) 583 |
| (d) 701 | (b) 501 |
| (e) None of these | (c) 552 |
| | (d) 427 |
| (10) 519, 743, 1029, 1359, 1763, 2239 | (e) None of these |
| (a) 1763 | |
| (b) 519 | (14) 150, 430, 1250, 3690, 10850, 32230 |
| (c) 1359 | (a) 430 |
| (d) 743 | (b) 10850 |
| (e) None of these | (c) 1250 |

- (d) 3690
(e) None of these

(15) 44, 270, 1329, 5325, 15991, 32007

- (a) 1329
(b) 15991
(c) 44
(d) 32007
(e) None of these

(16) 2400, 2419, 2455, 2510, 2599, 2695

- (a) 2599
(b) 2455
(c) 2419
(d) 2695
(e) None of these

(17) 300, 420, 550, 690, 840, 1015

- (a) 840
(b) 1015
(c) 550
(d) 300
(e) None of these

(18) 5000, 4970, 4934, 4834, 4654, 4360

- (a) 4934
(b) 4654
(c) 4970
(d) 4360
(e) None of these

(19) 930, 958, 1004, 1072, 1178, 1290

- (a) 958
(b) 1004
(c) 1290
(d) 1178
(e) None of these

(20) 210, 403, 787, 1560, 3073, 6115

- (a) 210
(b) 787
(c) 6115
(d) 3073
(e) None of these

Answers:

- (1) a
(2) b
(3) c
(4) d
(5) e
(6) a

(7) b

(8) c

(9) d

(10) e

(11) a

(12) b

(13) c

(14) d

(15) e

(16) a

(17) b

(18) c

(19) d

(20) e

Solutions:

(1) +38, +53, +70, +89, +110; second differences: +15, +17, +19, +21.

(2) $-(4^3 - 4)$, $-(5^3 - 5)$, $-(6^3 - 6)$, $-(7^3 - 7)$, $-(8^3 - 8)$.

(3) $\times 2 + 3$, $\times 2 + 6$, $\times 2 + 10$, $\times 2 + 15$, $\times 2 + 21$.

(4) $\div 2 + 11$, $\div 3 + 13$, $\div 4 + 17$, $\div 5 + 19$, $\div 6 + 23$.

(5) +44, +52, +68, +92, +124; second differences: +8, +16, +24, +32.

(6) $\times 2 + 14$, $\times 3 + 21$, $\times 4 + 28$, $\times 5 + 35$, $\times 6 + 42$.

(7) $-(4 \times 9)$, $-(5 \times 11)$, $-(6 \times 13)$, $-(7 \times 15)$, $-(8 \times 17)$.

(8) +6.25, +12.5, +25, +50, +100.

(9) $23^2 + 4$, $24^2 + 8$, $25^2 + 12$, $26^2 + 16$, $27^2 + 20$, $28^2 + 24$.

(10) $8^3 + 7$, $9^3 + 14$, $10^3 + 21$, $11^3 + 28$, $12^3 + 35$, $13^3 + 42$.

(11) $+(3 \times 6)$, $+(5 \times 8)$, $+(7 \times 10)$, $+(9 \times 12)$, $+(11 \times 14)$.

(12) $\div 3 + 5$, $\div 3 + 10$, $\div 3 + 20$, $\div 3 + 40$, $\div 3 + 80$.

(13) $20^2 - 7$, $21^2 - 14$, $22^2 - 21$, $23^2 - 28$, $24^2 - 35$, $25^2 - 42$.

(14) $\times 3 - 20$, $\times 3 - 40$, $\times 3 - 80$, $\times 3 - 160$, $\times 3 - 320$.

(15) $\times 6 + 1^2$, $\times 5 + 2^2$, $\times 4 + 3^2$, $\times 3 + 4^2$, $\times 2 + 5^2$.

(16) +19, +36, +55, +78, +107; second differences: +17, +19, +23, +29.

(17) $\times 4/3 + 20$, $\times 5/4 + 25$, $\times 6/5 + 30$, $\times 7/6 + 35$, $\times 8/7 + 40$.

(18) $-(3^2 \times 2)$, $-(4^2 \times 3)$, $-(5^2 \times 4)$, $-(6^2 \times 5)$, $-(7^2 \times 6)$.

(19) +28, +46, +68, +94, +124; second differences: +18, +22, +26, +30.

(20) $\times 2 - 17$, $\times 2 - 19$, $\times 2 - 23$, $\times 2 - 29$, $\times 2 - 31$.

5. MISSING NUMBER SERIES

(1) 7825, 7812, 7780, ?, 7581, 7360

- (a) 7709
- (b) 7711
- (c) 7713
- (d) 7715
- (e) None of these

(2) 8, 25, 85, ?, 1767, 10615

- (a) 347
- (b) 349
- (c) 351
- (d) 353
- (e) None of these

(3) 46, 58, 82, ?, 182, 266

- (a) 120
- (b) 122
- (c) 124
- (d) 126
- (e) None of these

(4) 112, 143, 262, ?, 702, 1149

- (a) 413
- (b) 415
- (c) 417

(d) 419

(e) None of these

(5) 4196, 2109, ?, 185, 42, 10

- (a) 708
- (b) 710
- (c) 712
- (d) 714

(e) None of these

(6) 65.2, 69.6, 78.4, ?, 109.2, 131.2

- (a) 91.6
- (b) 90.6
- (c) 92.4
- (d) 91.2

(e) None of these

(7) 72, 90, 120, ?, 246, 372

- (a) 166
- (b) 168
- (c) 170
- (d) 172

(e) None of these

(8) 39, 69, 122, ?, 402, 755

- (a) 219
- (b) 221
- (c) 217

- (d) 215
(e) None of these
- (9) 196, 206, 236, ?, 434, 656
(a) 300
(b) 302
(c) 306
(d) 304
(e) None of these
- (10) 2400, 480, 1920, ?, 2240, 280
(a) 320
(b) 310
(c) 315
(d) 325
(e) None of these
- (11) 4, 15, 54, ?, 1148, 6903
(a) 221
(b) 223
(c) 225
(d) 227
(e) None of these
- (12) 980, 970, 949, ?, 858, 780
(a) 911
(b) 913
(c) 915
(d) 917
(e) None of these
- (13) 312, 323, 350, ?, 484, 607
(a) 401
(b) 399
(c) 403
(d) 397
(e) None of these
- (14) 28, 72, 144, ?, 396, 588

- (a) 248
(b) 250
(c) 252
(d) 246
(e) None of these
- (15) 2200, 2224, 2296, ?, 2756, 3260
(a) 2452
(b) 2454
(c) 2458
(d) 2456
(e) None of these
- (16) 88, 196, 408, ?, 1664, 3332
(a) 828
(b) 826
(c) 830
(d) 824
(e) None of these
- (17) 80, 95, 130, ?, 292, 435
(a) 189
(b) 191
(c) 195
(d) 193
(e) None of these
- (18) 240, 254, 296, ?, 800, 1934
(a) 418
(b) 420
(c) 422
(d) 424
(e) None of these
- (19) 3520, 3507, 3482, ?, 3380, 3295
(a) 3441
(b) 3439
(c) 3443

- (d) 3437
 (e) None of these
 (20) 9, 18, 37, 76, ?, 314
 (a) 151
 (b) 153
 (c) 155
 (d) 157
 (e) None of these

Answers:

- (1) b
 (2) c
 (3) b
 (4) d
 (5) c
 (6) a
 (7) b
 (8) a
 (9) d
 (10) a
 (11) d
 (12) b
 (13) a
 (14) b
 (15) d
 (16) a
 (17) d
 (18) c
 (19) a
 (20) c

Solutions:

- (1) $-(2^3 + 5)$, $-(3^3 + 5)$, $-(4^3 + 5)$,
 $-(5^3 + 5)$, $-(6^3 + 5)$

$$(2) \times 2 + 9, \times 3 + 10, \times 4 + 11, \times 5 + 12, \times 6 + 13$$

$$(3) +2 \times 6, +3 \times 8, +4 \times 10, +5 \times 12, +6 \times 14$$

$$(4) \text{ Sum of previous two numbers } +7, +14, +21, +28$$

$$(5) \div 2 + 11, \div 3 + 9, \div 4 + 7, \div 5 + 5, \div 6 + 3$$

$$(6) +4.4, +8.8, +13.2, +17.6, +22$$

$$(7) +18, +30, +48, +78, +126$$

Each difference is the sum of the previous two differences.

$$(8) \times 2 - 3^2, \times 2 - 4^2, \times 2 - 5^2, \times 2 - 6^2, \times 2 - 7^2$$

$$(9) +2^3 + 2, +3^3 + 3, +4^3 + 4, +5^3 + 5, +6^3 + 6$$

$$(10) \div 5, \times 4, \div 6, \times 7, \div 8$$

$$(11) \times 2 + 7, \times 3 + 9, \times 4 + 11, \times 5 + 13, \times 6 + 15$$

$$(12) -5 \times 2, -7 \times 3, -9 \times 4, -11 \times 5, -13 \times 6$$

$$(13) +11, +27, +51, +83, +123$$

$$\text{Second difference: } +16, +24, +32, +40$$

$$(14) \text{ Terms are: } n^3 + 5n^2, \text{ for } n = 2, 3, 4, 5, 6, 7$$

$$(15) +2^2 \times 6, +3^2 \times 8, +4^2 \times 10, +5^2 \times 12, +6^2 \times 14$$

$$(16) \times 2 + 20, \times 2 + 16, \times 2 + 12, \times 2 + 8, \times 2 + 4$$

(17) $+(4^2 - 1)$, $+(6^2 - 1)$, $+(8^2 - 1)$,
 $+(10^2 - 1)$, $+(12^2 - 1)$

(18) $+14$, $+42$, $+126$, $+378$, $+1134$

Each difference is multiplied by 3.

(19) $-(2^2 + 3^2)$, $-(3^2 + 4^2)$, $-(4^2 + 5^2)$,
 $-(5^2 + 6^2)$, $-(6^2 + 7^2)$

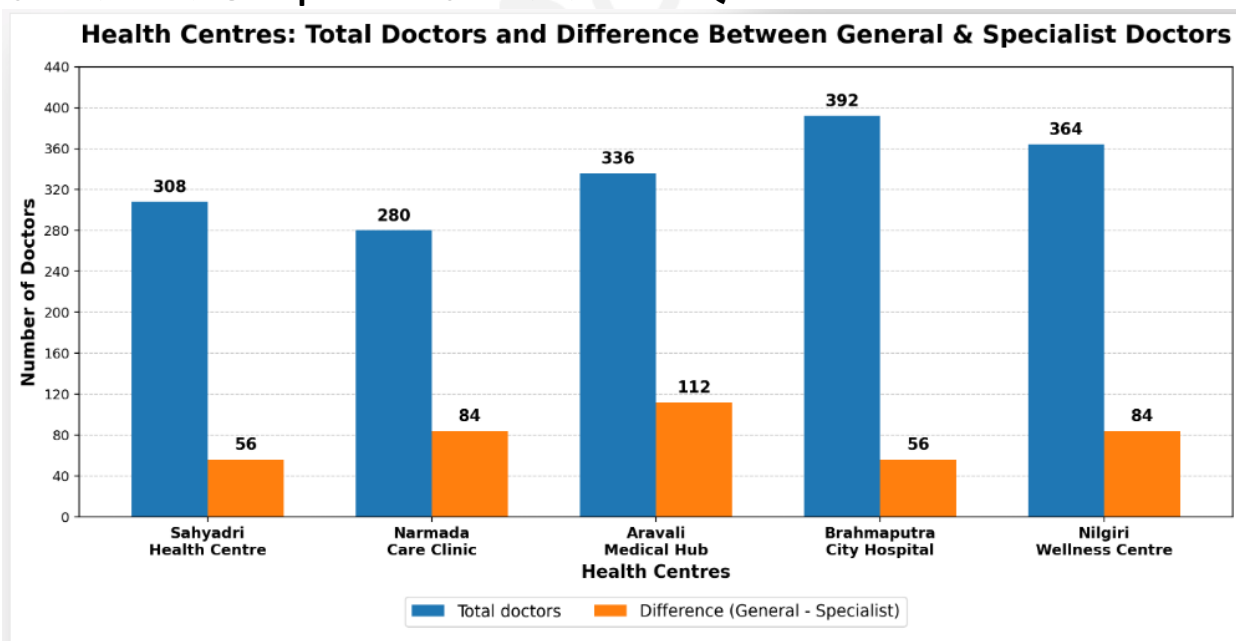
(20) $\times 2 + 0$, $\times 2 + 1$, $\times 2 + 2$, $\times 2 + 3$,
 $\times 2 + 4$

CHECKLIST
BY
AASHISH
ARORA

6. DATA INTERPRETATION

Set 1: The bar graph shows the total number of doctors and the table shows the difference between General doctors and Specialist doctors in five different health centres. Total doctors = General doctors + Specialist doctors. In each health centre, the number of General doctors is more than the number of Specialist doctors.

दंड आलेख पाँच अलग-अलग स्वास्थ्य केंद्रों में डॉक्टरों की कुल संख्या दर्शाता है और तालिका General डॉक्टरों तथा Specialist डॉक्टरों के बीच का अंतर दर्शाती है। कुल डॉक्टर = General डॉक्टर + Specialist डॉक्टर। प्रत्येक स्वास्थ्य केंद्र में General डॉक्टरों की संख्या Specialist डॉक्टरों से अधिक है।



Q1: Find the difference between the average number of General doctors in Narmada Care Clinic and Brahmaputra City Hospital and the average number of Specialist doctors in Sahyadri Health Centre and Nilgiri Wellness Centre.

नर्मदा केयर क्लिनिक और ब्रह्मपुत्र सिटी हॉस्पिटल में General डॉक्टरों की औसत संख्या तथा सह्याद्री हेल्थ सेंटर और नीलगिरि वेलनेस सेंटर में Specialist डॉक्टरों की औसत संख्या के बीच अंतर ज्ञात कीजिए।

- (A) 56
- (B) 63
- (C) 84

(D) 70

(E) 77

Q2: In Sahyadri Health Centre, all doctors are either Consultants or Resident doctors. The ratio of Consultants to Resident doctors is 7:4. Female Consultants are $16\frac{2}{3}\%$ of Male Consultants. Find the difference between Male Consultants and total Resident doctors in Sahyadri Health Centre.

सह्याद्री हेल्थ सेंटर में सभी डॉक्टर या तो Consultants हैं या Resident doctors हैं। Consultants और Resident doctors का अनुपात 7:4 है। Female Consultants, Male Consultants का $16\frac{2}{3}\%$ हैं। सह्याद्री हेल्थ सेंटर में Male Consultants और कुल Resident doctors के बीच अंतर ज्ञात कीजिए।

(A) 56

(B) 49

(C) 63

(D) 70

(E) 42

Q3: In a new health centre Triveni, the total number of doctors is equal to the average number of total doctors in Sahyadri Health Centre, Aravali Medical Hub and Nilgiri Wellness Centre. If the ratio of General doctors in Sahyadri Health Centre and Triveni is 1:1, then find the ratio of Specialist doctors in Triveni to General doctors in Nilgiri Wellness Centre. If the required ratio is a:b, find a + b.

एक नए स्वास्थ्य केंद्र त्रिवेणी में कुल डॉक्टरों की संख्या, सह्याद्री हेल्थ सेंटर, अरावली मेडिकल हब और नीलगिरि वेलनेस सेंटर में कुल डॉक्टरों की औसत संख्या के बराबर है। यदि सह्याद्री हेल्थ सेंटर और त्रिवेणी में General डॉक्टरों का अनुपात 1:1 है, तो त्रिवेणी में Specialist डॉक्टरों और नीलगिरि वेलनेस सेंटर में General डॉक्टरों का अनुपात ज्ञात कीजिए। यदि आवश्यक अनुपात a:b है, तो a + b ज्ञात कीजिए।

(A) 25

(B) 23

(C) 31

(D) 29

(E) 27

Q4: Find the difference between the total number of General doctors and the total number of Specialist doctors in all five health centres together.

सभी पाँच स्वास्थ्य केंद्रों में मिलाकर General डॉक्टरों की कुल संख्या और Specialist डॉक्टरों की कुल संख्या के बीच अंतर ज्ञात कीजिए।

- (A) 364
(B) 378
(C) 392
(D) 406
(E) 420

Q5: In Aravali Medical Hub, the number of female General doctors is 154. The number of male General doctors is 42 more than the number of female Specialist doctors. Male Specialist doctors in Aravali Medical Hub are what percentage less than total Specialist doctors in Brahmaputra City Hospital?

अरावली मेडिकल हब में Female General डॉक्टरों की संख्या 154 है। Male General डॉक्टरों की संख्या Female Specialist डॉक्टरों से 42 अधिक है। अरावली मेडिकल हब में Male Specialist डॉक्टर, ब्रह्मपुत्र सिटी हॉस्पिटल में कुल Specialist डॉक्टरों से कितने प्रतिशत कम हैं?

- (A) 25%
(B) 50%
(C) $37\frac{1}{2}\%$
(D) $33\frac{1}{3}\%$
(E) 40%

Solutions:

Health Centre	Total	Difference	General Doctors	Specialist Doctors	Formula Used
Sahyadri Health Centre	308	56	182	126	$(308 + 56)/2, (308 - 56)/2$
Narmada Care Clinic	280	84	182	98	$(280 + 84)/2, (280 - 84)/2$
Aravali Medical Hub	336	112	224	112	$(336 + 112)/2, (336 - 112)/2$
Brahmaputra City Hospital	392	56	224	168	$(392 + 56)/2, (392 - 56)/2$
Nilgiri Wellness Centre	364	84	224	140	$(364 + 84)/2, (364 - 84)/2$

1. Option D

General doctors in Narmada Care Clinic = $(280 + 84) / 2 = 182$

General doctors in Brahmaputra City Hospital = $(392 + 56) / 2 = 224$

Average General doctors = $(182 + 224) / 2 = 203$

Specialist doctors in Sahyadri Health Centre = $(308 - 56) / 2 = 126$

Specialist doctors in Nilgiri Wellness Centre = $(364 - 84) / 2 = 140$

Average Specialist doctors = $(126 + 140) / 2 = 133$

Difference = $203 - 133 = 70$

2. Option A

Total doctors in Sahyadri Health Centre = 308

Consultants : Resident doctors = 7:4

Total parts = 11

1 part = $308 / 11 = 28$

Consultants = $7 \times 28 = 196$

Resident doctors = $4 \times 28 = 112$

Female Consultants = $16 \frac{2}{3}\%$ of Male Consultants = $\frac{1}{6}$ of Male Consultants

Male Consultants + Female Consultants = 196

Male Consultants + $\frac{1}{6}$ Male Consultants = 196

$\frac{7}{6}$ Male Consultants = 196

Male Consultants = 168

Difference = $168 - 112 = 56$

3. Option E

Total doctors in Triveni = $(308 + 336 + 364) / 3 = 336$

General doctors in Sahyadri Health Centre = $(308 + 56) / 2 = 182$

General doctors in Triveni = 182

Specialist doctors in Triveni = $336 - 182 = 154$

General doctors in Nilgiri Wellness Centre = $(364 + 84) / 2 = 224$

Required ratio = $154:224 = 11:16$

$a + b = 11 + 16 = 27$

4. Option C

Difference between General and Specialist doctors in all centres = $56 + 84 + 112 + 56 + 84$

Difference = 392

5. Option B

General doctors in Aravali Medical Hub = $(336 + 112) / 2 = 224$

Specialist doctors in Aravali Medical Hub = $(336 - 112) / 2 = 112$

Female General doctors in Aravali Medical Hub = 154

Male General doctors in Aravali Medical Hub = $224 - 154 = 70$

Male General doctors = Female Specialist doctors + 42

Female Specialist doctors = $70 - 42 = 28$

Male Specialist doctors = $112 - 28 = 84$

Specialist doctors in Brahmaputra City Hospital = $(392 - 56) / 2 = 168$

Percentage less = $(168 - 84) / 168 \times 100 = 50\%$

Set 2: The table shows the total number of parcels sent by bike and van together in four different courier hubs. It also shows the number of parcels sent by drone and the number of parcels sent by van. Each parcel was sent through only one mode: bike, van or drone.

तालिका चार अलग-अलग कूरियर हबों में बाइक और वैन से मिलाकर भेजे गए पार्सलों की कुल संख्या दर्शाती है। यह ड्रोन से भेजे गए पार्सलों की संख्या और वैन से भेजे गए पार्सलों की संख्या भी दर्शाती है। प्रत्येक पार्सल केवल एक माध्यम से भेजा गया: बाइक, वैन या ड्रोन।

Courier Hub कूरियर हब	Bike + Van Parcels बाइक + वैन पार्सल	Drone Parcels ड्रोन पार्सल	Van Parcels वैन पार्सल
Latur Hub लातूर हब	780	520	325
Bikaner Hub बीकानेर हब	845	650	390
Panipat Hub पानीपत हब	910	455	260
Raigarh Hub रायगढ़ हब	1144	715	624

Q1: The total number of parcels sent by drone from Bikaner Hub is what percentage of the total number of parcels sent by bike from Raigarh Hub?
बीकानेर हब से ड्रोन द्वारा भेजे गए कुल पार्सल, रायगढ़ हब से बाइक द्वारा भेजे गए कुल पार्सलों का कितने प्रतिशत हैं?

- (A) 100%
(B) 150%
(C) 75%
(D) 125%
(E) 80%

Q2: The number of parcels sent by drone from Panipat Hub is how many more or less than the number of parcels sent by van from Latur Hub?
पानीपत हब से ड्रोन द्वारा भेजे गए पार्सल, लातूर हब से वैन द्वारा भेजे गए पार्सलों से कितने अधिक या कम हैं?

- (A) 117
(B) 130
(C) 156
(D) 104
(E) 143

Q3: If the number of parcels sent through express courier from Vellore Hub is $\frac{5}{7}$ of the number of parcels sent by bike from Bikaner Hub, then find the average number of parcels sent through express courier from Vellore Hub and parcels sent by drone from Raigarh Hub.

यदि वेल्लोर हब से एक्सप्रेस कूरियर द्वारा भेजे गए पार्सलों की संख्या, बीकानेर हब से बाइक द्वारा भेजे गए पार्सलों की संख्या का $\frac{5}{7}$ है, तो वेल्लोर हब से एक्सप्रेस कूरियर द्वारा भेजे गए पार्सलों और रायगढ़ हब से ड्रोन द्वारा भेजे गए पार्सलों की औसत संख्या ज्ञात कीजिए।

- (A) 585
- (B) 455
- (C) 540
- (D) 495
- (E) 520

Q4: If the ratio of parcels sent by bike from Latur Hub to parcels sent by van from Panipat Hub is $P:Q$, then find $P + Q$.

यदि लातूर हब से बाइक द्वारा भेजे गए पार्सलों और पानीपत हब से वैन द्वारा भेजे गए पार्सलों का अनुपात $P:Q$ है, तो $P + Q$ ज्ञात कीजिए।

- (A) 11
- (B) 13
- (C) 15
- (D) 17
- (E) 19

Q5: The number of parcels packed manually in Ernakulam Hub is 25% less than the number of parcels sent by van from Raigarh Hub. Find the sum of manually packed parcels in Ernakulam Hub and parcels sent by bike from Bikaner Hub.

एर्नाकुलम हब में मैनुअल रूप से पैक किए गए पार्सलों की संख्या, रायगढ़ हब से वैन द्वारा भेजे गए पार्सलों से 25% कम है। एर्नाकुलम हब में मैनुअल रूप से पैक किए गए पार्सलों और बीकानेर हब से बाइक द्वारा भेजे गए पार्सलों का योग ज्ञात कीजिए।

- (A) 871
- (B) 910
- (C) 923
- (D) 936
- (E) 949

Solutions:

Courier Hub कूरियर हब	Bike Parcels बाइक पार्सल	Van Parcels वैन पार्सल	Drone Parcels ड्रोन पार्सल
Latur Hub लातूर हब	455	325	520
Bikaner Hub बीकानेर हब	455	390	650
Panipat Hub पानीपत हब	650	260	455
Raigarh Hub रायगढ़ हब	520	624	715

Common values:

Latur Hub: Bike = $780 - 325 = 455$

Bikaner Hub: Bike = $845 - 390 = 455$

Panipat Hub: Bike = $910 - 260 = 650$

Raigarh Hub: Bike = $1144 - 624 = 520$

1. Option D

Drone parcels from Bikaner Hub = 650

Bike parcels from Raigarh Hub = 520

Required percentage = $650/520 \times 100 = 125\%$

2. Option B

Drone parcels from Panipat Hub = 455

Van parcels from Latur Hub = 325

Required difference = $455 - 325 = 130$

3. Option E

Bike parcels from Bikaner Hub = 455

Express courier parcels from Vellore Hub = $5/7 \times 455 = 325$

Drone parcels from Raigarh Hub = 715

Required average = $(325 + 715)/2 = 520$

4. Option A

Bike parcels from Latur Hub = 455

Van parcels from Panipat Hub = 260

Ratio = $455:260 = 7:4$

$P + Q = 7 + 4 = 11$

5. Option C

Van parcels from Raigarh Hub = 624

Manually packed parcels in Ernakulam Hub = 75% of 624 = 468

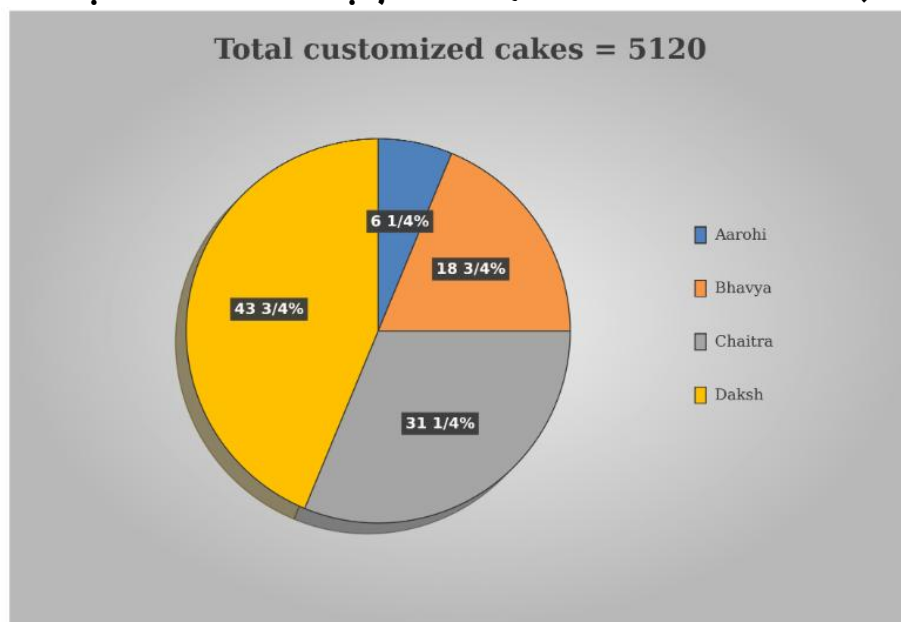
Bike parcels from Bikaner Hub = 455

Required sum = 468 + 455 = 923

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Set 3: The pie chart shows the percentage distribution of total customized cakes, chocolate and fruit cakes together, prepared by four different bakeries: AaroHi, Bhavya, Chaitra and Daksh. The total number of cakes prepared by all four bakeries together is 5120. Total cakes prepared by a bakery = Chocolate cakes + Fruit cakes. The information table gives the data required to calculate chocolate and fruit cakes for each bakery. Read the data carefully and answer the following questions.

पाई चार्ट चार अलग-अलग बेकरी, आरोही, भाव्या, चैत्रा और दक्ष द्वारा बनाए गए कुल कस्टमाइज्ड केक, यानी चॉकलेट और फ्रूट केक मिलाकर, का प्रतिशत वितरण दर्शाता है। सभी चार बेकरी द्वारा मिलाकर बनाए गए कुल केक की संख्या 5120 है। किसी बेकरी द्वारा बनाए गए कुल केक = चॉकलेट केक + फ्रूट केक। सूचना तालिका प्रत्येक बेकरी के चॉकलेट और फ्रूट केक ज्ञात करने के लिए आवश्यक आँकड़े देती है। आँकड़ों को ध्यान से पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



Bakery	Information
AaroHi	The number of chocolate cakes is 37.5% of total cakes.
Bhavya	Fruit cakes prepared by AaroHi are equal to chocolate cakes prepared by Bhavya.
Chaitra	Chocolate cakes prepared are 32 more than fruit cakes.
Daksh	The ratio of chocolate cakes to fruit cakes is 5:9.

Q1: If 25% of the total fruit cakes prepared by AaroHi are mango cakes and the ratio of mango cakes to pineapple cakes is 2:5, then find the difference between apple fruit cakes prepared by AaroHi and total chocolate cakes prepared by Daksh. Fruit cakes = Mango cakes + Pineapple cakes + Apple cakes.

यदि आरोही द्वारा बनाए गए कुल फ्रूट केक में से 25% मैंगो केक हैं और मैंगो केक तथा

पाइनएप्पल केक का अनुपात 2:5 है, तो आरोही द्वारा बनाए गए एप्पल फ्रूट केक और दक्ष द्वारा बनाए गए कुल चॉकलेट केक के बीच का अंतर ज्ञात कीजिए। फ्रूट केक = मैंगो केक + पाइनएप्पल केक + एप्पल केक।

- (A) 725
- (B) 750
- (C) 800
- (D) 775
- (E) 825

Q2: Find the sum of the difference between chocolate and fruit cakes prepared by Bhavya and the sum of fruit cakes prepared by Chaitra and Daksh.

भाव्या द्वारा बनाए गए चॉकलेट और फ्रूट केक के बीच के अंतर तथा चैत्रा और दक्ष द्वारा बनाए गए फ्रूट केक के योग का कुल योग ज्ञात कीजिए।

- (A) 2784
- (B) 2640
- (C) 2704
- (D) 2864
- (E) 2928

Q3: The total cakes prepared by a new bakery, Esha, are 20% less than the total cakes prepared by Bhavya and Chaitra together. The ratio of chocolate cakes prepared by Aarohi to chocolate cakes prepared by Esha is 3:16. Find the number of fruit cakes prepared by Esha.

एक नई बेकरी, ईशा, द्वारा बनाए गए कुल केक, भाव्या और चैत्रा द्वारा मिलाकर बनाए गए कुल केक से 20% कम हैं। आरोही द्वारा बनाए गए चॉकलेट केक और ईशा द्वारा बनाए गए चॉकलेट केक का अनुपात 3:16 है। ईशा द्वारा बनाए गए फ्रूट केक की संख्या ज्ञात कीजिए।

- (A) 1344
- (B) 1472
- (C) 1536
- (D) 1280
- (E) 1408

Q4: Find the ratio of total chocolate cakes prepared by Aarohi and Bhavya together to total fruit cakes prepared by Chaitra.

आरोही और भाव्या द्वारा मिलाकर बनाए गए कुल चॉकलेट केक का चैत्रा द्वारा बनाए गए कुल फ्रूट केक से अनुपात ज्ञात कीजिए।

- (A) 19:47

- (B) 20:49
 (C) 25:49
 (D) 49:20
 (E) 40:49

Q5: Fruit cakes prepared by Bhavya are what percentage of chocolate cakes prepared by Daksh?

भाव्या द्वारा बनाए गए फ्रूट केक, दक्ष द्वारा बनाए गए चॉकलेट केक का कितना प्रतिशत हैं?

- (A) 80%
 (B) 90%
 (C) 95%
 (D) 100%
 (E) 85%

Solutions:

Common Calculation:

Total pie-chart value = $16 + 48 + 80 + 112 = 256$

1 pie-chart value = $5120 / 256 = 20$ cakes

Aarohi chocolate = 37.5% of $320 = 120$; fruit = 200

Bhavya chocolate = Aarohi fruit = 200 ; fruit = 760

Chaitra: fruit = a , chocolate = $a + 32$, so $a + a + 32 = 1600 \Rightarrow$ fruit = 784 , chocolate = 816

Daksh chocolate : fruit = $5 : 9 \Rightarrow$ chocolate = 800 , fruit = 1440

Bakery	Total Cakes	Chocolate Cakes	Fruit Cakes
Aarohi	320	120	200
Bhavya	960	200	760
Chaitra	1600	816	784
Daksh	2240	800	1440

Common Calculation:

Total cakes = 5120

Total pie-chart value = $16 + 48 + 80 + 112 = 256$

Aarohi total cakes = $16/256 \times 5120 = 320$

Bhavya total cakes = $48/256 \times 5120 = 960$

Chaitra total cakes = $80/256 \times 5120 = 1600$

Daksh total cakes = $112/256 \times 5120 = 2240$

Aarohi chocolate cakes = 37.5% of $320 = 120$

Aarohi fruit cakes = $320 - 120 = 200$

Bhavya chocolate cakes = Aarohi fruit cakes = 200

Bhavya fruit cakes = $960 - 200 = 760$

Let Chaitra fruit cakes = a

Chaitra chocolate cakes = $a + 32$

$a + a + 32 = 1600$

$a = 784$

Chaitra fruit cakes = 784 and chocolate cakes = 816

Daksh chocolate:fruit = 5:9

Daksh chocolate cakes = $2240 \times \frac{5}{14} = 800$

Daksh fruit cakes = $2240 \times \frac{9}{14} = 1440$

1. Option D

Aarohi fruit cakes = 200

Mango cakes = 25% of 200 = 50

Mango:Pineapple = 2:5

Pineapple cakes = $50 \times \frac{5}{2} = 125$

Apple fruit cakes = $200 - 50 - 125 = 25$

Daksh chocolate cakes = 800

Required difference = $800 - 25 = 775$

2. Option A

Bhavya chocolate cakes = 200

Bhavya fruit cakes = 760

Difference = $760 - 200 = 560$

Chaitra + Daksh fruit cakes = $784 + 1440 = 2224$

Required sum = $560 + 2224 = 2784$

3. Option E

Bhavya + Chaitra total cakes = $960 + 1600 = 2560$

Esha total cakes = 80% of 2560 = 2048

Aarohi:Esha chocolate cakes = 3:16

Esha chocolate cakes = $120 \times \frac{16}{3} = 640$

Esha fruit cakes = $2048 - 640 = 1408$

4. Option B

Aarohi + Bhavya chocolate cakes = $120 + 200 = 320$

Chaitra fruit cakes = 784

Required ratio = $320:784 = 20:49$

5. Option C

Bhavya fruit cakes = 760

Daksh chocolate cakes = 800

Required percentage = $760/800 \times 100 = 95\%$

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Set 4: Three handmade craft stalls, Nakul, Tanvi and Harit, sold three different types of craft kits: Origami Kits, Painting Kits and Clay Kits. The numbers of Clay Kits and Painting Kits sold by Nakul are 50% more and 25% more than the number of Origami Kits sold by Nakul, respectively. The number of Origami Kits sold by Tanvi is 119 less than the number of Painting Kits sold by Nakul. The number of Clay Kits sold by Tanvi is equal to the number of Origami Kits sold by Nakul. The ratio of Painting Kits sold by Harit to Painting Kits sold by Nakul is 8 : 5. The number of Clay Kits sold by Harit is equal to the number of Painting Kits sold by Tanvi. The number of Origami Kits sold by Harit is 748. The average number of all three types of craft kits sold by Nakul is 510. The average number of Clay Kits sold by all three stalls together is 680.

तीन हस्तनिर्मित क्राफ्ट स्टॉल, नकुल, तन्वी और हरित, ने तीन अलग-अलग प्रकार की क्राफ्ट किट बेचीं: ओरिगामी किट, पेंटिंग किट और क्ले किट। नकुल द्वारा बेची गई क्ले किट और पेंटिंग किट की संख्या, नकुल द्वारा बेची गई ओरिगामी किट की संख्या से क्रमशः 50% अधिक और 25% अधिक है। तन्वी द्वारा बेची गई ओरिगामी किट की संख्या, नकुल द्वारा बेची गई पेंटिंग किट की संख्या से 119 कम है। तन्वी द्वारा बेची गई क्ले किट की संख्या, नकुल द्वारा बेची गई ओरिगामी किट की संख्या के बराबर है। हरित और नकुल द्वारा बेची गई पेंटिंग किट की संख्या का अनुपात 8 : 5 है। हरित द्वारा बेची गई क्ले किट की संख्या, तन्वी द्वारा बेची गई पेंटिंग किट की संख्या के बराबर है। हरित द्वारा बेची गई ओरिगामी किट की संख्या 748 है। नकुल द्वारा बेची गई तीनों प्रकार की क्राफ्ट किट की औसत संख्या 510 है। तीनों स्टॉल द्वारा बेची गई क्ले किट की औसत संख्या 680 है।

Q1: Find the difference between the total number of Clay Kits sold by Nakul and Tanvi together and the total number of Origami Kits sold by Tanvi and Harit together.

नकुल और तन्वी द्वारा मिलाकर बेची गई क्ले किट की कुल संख्या तथा तन्वी और हरित द्वारा मिलाकर बेची गई ओरिगामी किट की कुल संख्या के बीच का अंतर ज्ञात कीजिए।

- (A) 102
- (B) 119
- (C) 136
- (D) 153
- (E) 170

Q2: The number of Painting Kits sold by Harit is what percentage more than the number of Clay Kits sold by Tanvi?

हरित द्वारा बेची गई पेंटिंग किट की संख्या, तन्वी द्वारा बेची गई क्ले किट की संख्या से कितने

प्रतिशत अधिक है?

- (A) 75% more
- (B) 125% more
- (C) 50% more
- (D) 100% more
- (E) 150% more

Q3: If the simplified ratio of Painting Kits sold by Nakul to Origami Kits sold by Harit is $a : b$, then find the value of $a + b$.

यदि नकुल द्वारा बेची गई पेंटिंग किट और हरित द्वारा बेची गई ओरिगामी किट का सरलतम अनुपात $a : b$ है, तो $a + b$ का मान ज्ञात कीजिए।

- (A) 37
- (B) 35
- (C) 39
- (D) 41
- (E) 43

Q4: Find the average number of Painting Kits sold by all three stalls.

तीनों स्टॉल द्वारा बेची गई पेंटिंग किट की औसत संख्या ज्ञात कीजिए।

- (A) 765
- (B) 799
- (C) 748
- (D) 816
- (E) 782

Q5: Find 40% of the total number of craft kits sold by Nakul.

नकुल द्वारा बेची गई कुल क्राफ्ट किट का 40% ज्ञात कीजिए।

- (A) 646
- (B) 578
- (C) 612
- (D) 629
- (E) 595

Solutions:

Stall	Origami Kits	Painting Kits	Clay Kits	Total Kits
Nakul	408	510	612	1530
Tanvi	391	1020	408	1819
Harit	748	816	1020	2584
Total	1547	2346	2040	5933

1. Option B

Clay Kits sold by Nakul and Tanvi together = $612 + 408$
 $= 1020$

Origami Kits sold by Tanvi and Harit together = $391 + 748$
 $= 1139$

Required difference = $1139 - 1020$
 $= 119$

2. Option D

Painting Kits sold by Harit = 816

Clay Kits sold by Tanvi = 408

Difference = $816 - 408$
 $= 408$

Required percentage more = $408 \div 408 \times 100$
 $= 100\%$ more

3. Option A

Painting Kits sold by Nakul = 510

Origami Kits sold by Harit = 748

Required ratio = $510 : 748$

$= 15 : 22$

$a + b = 15 + 22$

$= 37$

4. Option E

Painting Kits sold by Nakul, Tanvi and Harit = 510, 1020 and 816

Total Painting Kits = $510 + 1020 + 816$

$= 2346$

Required average = $2346 \div 3$

$= 782$

5. Option C

Total craft kits sold by Nakul = 1530

Required value = 40% of 1530

= 612

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